

A Safety-Aware Path Smoothing and Turning-Maneuver Optimization Method for a Two-Wheeled Differential-Drive Mobile Robot

Intelligent Control, Robotics, and Automation — Autonomous Systems (AMR, AGV, UAV)

Abstract—Global planners in ROS 2/Nav2 return collision-free polylines with abrupt heading changes. For a two-wheeled differential-drive mobile robot, preserving these vertices can induce frequent in-place rotations and high curvature variation, whereas unconstrained smoothing may cut corners and reduce obstacle clearance. This study addresses footprint-safe path post-processing in narrow warehouse passages. We propose a path smoothing and turning-maneuver optimization method (PSTMO) that conditions the planner path and represents each retained corner either as an explicit in-place pivot or as a quintic Bézier transition that is geometrically G^2 -continuous with adjacent line segments. An adaptive trim/radius search generates local candidates; curvature, wheel-motion, and swept-footprint checks reject infeasible candidates; and dynamic programming selects a globally compatible sequence without overlap between adjacent transitions. Evaluation used 175 paired path records from seven representative scenarios and five global planners, comparing Raw, three stock Nav2 smoothers (Simple, Savitzky–Golay, and Constrained), and PSTMO with identical raw paths within each group. PSTMO, Savitzky–Golay, and Constrained succeeded in all 35 cases, while Simple succeeded in 34. On the 34 groups successful for every method, PSTMO achieved the lowest mean translational curvature energy (4.51 m^{-1} versus $9.42\text{--}50.21 \text{ m}^{-1}$) and shortest mean path (10.19 m versus 10.24–10.33 m); Constrained provided the largest mean minimum footprint clearance (0.213 m versus PSTMO's 0.176 m). No successful output contained footprint-collision samples. Six closed-loop Gazebo trials forming three Raw–PSTMO pairs also completed without collision-monitor intervention; PSTMO reduced mean execution time from 61.47 to 59.02 s and mean ground-truth tracking RMSE from 2.85 to 2.81 cm. The results indicate that PSTMO provides footprint-aware path smoothing while retaining in-place rotations where continuous cornering is infeasible within this reduced evaluation. Repeated trials, broader scenario coverage, and hardware experiments remain future work.

Keywords—Two-wheeled mobile robot; differential drive; path smoothing; maneuver optimization; swept-footprint safety; ROS 2/Nav2.